

AgriNova

Crop residue as income. Not smoke.

A live marketplace for farmers, biomass buyers, and agriculture officers in Punjab - with Kisan AI, soil cards, dual carbon credits, and a government pollution heatmap.

- Live: <https://agrinova-ochre.vercel.app>
- Four role-separated desks from one login (logout to switch)
- Punjab case study (Patiala-Sangrur) with lived-in demo data
- PDF + in-app slides for judges who will not install Node

INSIDE THIS DECK

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- 02 How the system is built
- 03 User journey
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- 05 Product snapshots
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- 07 Burn vs sell calculator
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- 09 Field & climate impact
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The harvest window forces a burn default

After pa
exist in t

Air

Peak November PM2.5 episodes show crop-fire share around 30-40% in public literature. District officers lack a utilised-vs-burned view.

Farmer

Burning earns Rs 0 and risks NGT fines. Smallholders cannot hire a baler alone or reach mill WhatsApp groups.

Buyer

Biomass plants need moisture-qualified straw (often $\leq 15\%$). Informal brokers skip 2-acre farms.

Gap

No shared desk that turns a mapped field into a matched offer, a pickup, and a credit - in one session.

One platform. Four desks. No mixed profiles.

Farmer

Plot map on OpenStreetMap, burn-vs-sell calculator, residue listing wizard, match score, pickup, credits, Kisan AI, soil card, learn farming.

Buyer

Procurement inbox, moisture specs, RFQs, dual credits, indicative green-tax / CSR simulator.

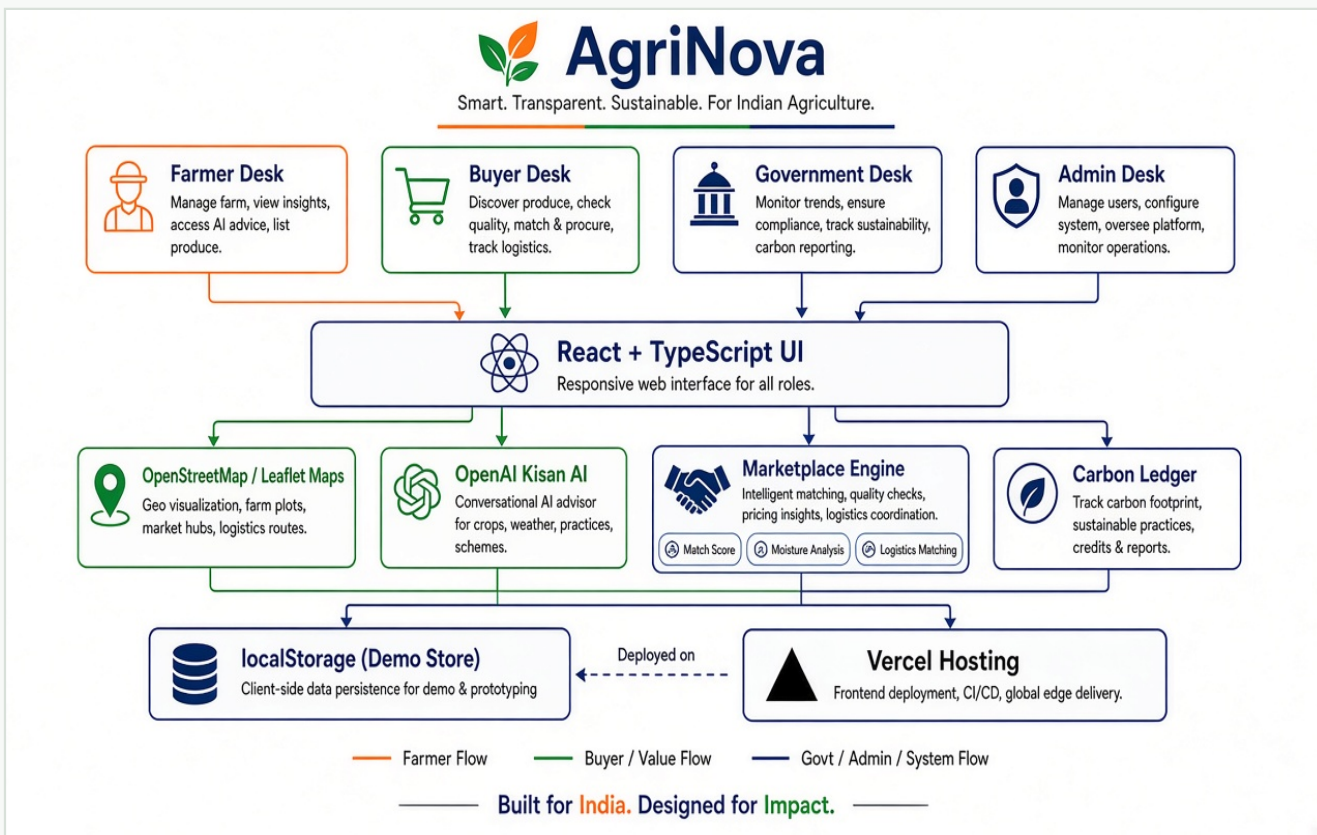
Government

Department-style portal: pollution heatmap, operational clusters, utilisation slider, savings table, Policy AI.

Admin

Transaction pipeline, carbon ledger, MRV queue for the demo cluster.

Architecture of the running product



Four desks, one engine

Roles sit on a React + TypeScript UI. Maps use Leaflet/OSM. Matching is rule-based and moisture-aware. Kisan AI and Policy AI call OpenAI when a key is present, else researched fallbacks. Marketplace state lives in the browser (localStorage) with lived-in seed data so a booth demo is never empty. Hosting is Vercel.

What the product is built on

Interface

React 19, TypeScript, Vite, Tailwind. White canvas with saffron-white-green tricolor. Role-scoped navigation and toast feedback.

Maps

Leaflet + OpenStreetMap. Farmers draw field corners in lat/lng; acres from geodesic area. Government heat and operational markers share the same basemap.

Marketplace

Classify residue, value it, rank pathways, Match Score (distance, demand, moisture, price, pathway). Transaction state machine through pickup and complete.

Decision tool

Burn-vs-sell calculator: acres, price, moisture, baling cost -> net income, NGT fine risk, credits, tCO₂e, top buyer match.

AI

OpenAI gpt-4o-mini with role system prompts. Voice input (browser speech recognition) and spoken replies. Soil card injected into farmer context.

Carbon

Avoided tCO₂e on completed sales. Farmer wallet (redeem seed/fertiliser). Buyer procurement credits feeding an indicative Rs/tCO₂e simulator.

Data

Punjab 25-farm pack + West Bengal comparison pack. Lived-in seed: Simran already sold; GreenPower RFQ is open.

From login to credits - and the officer path

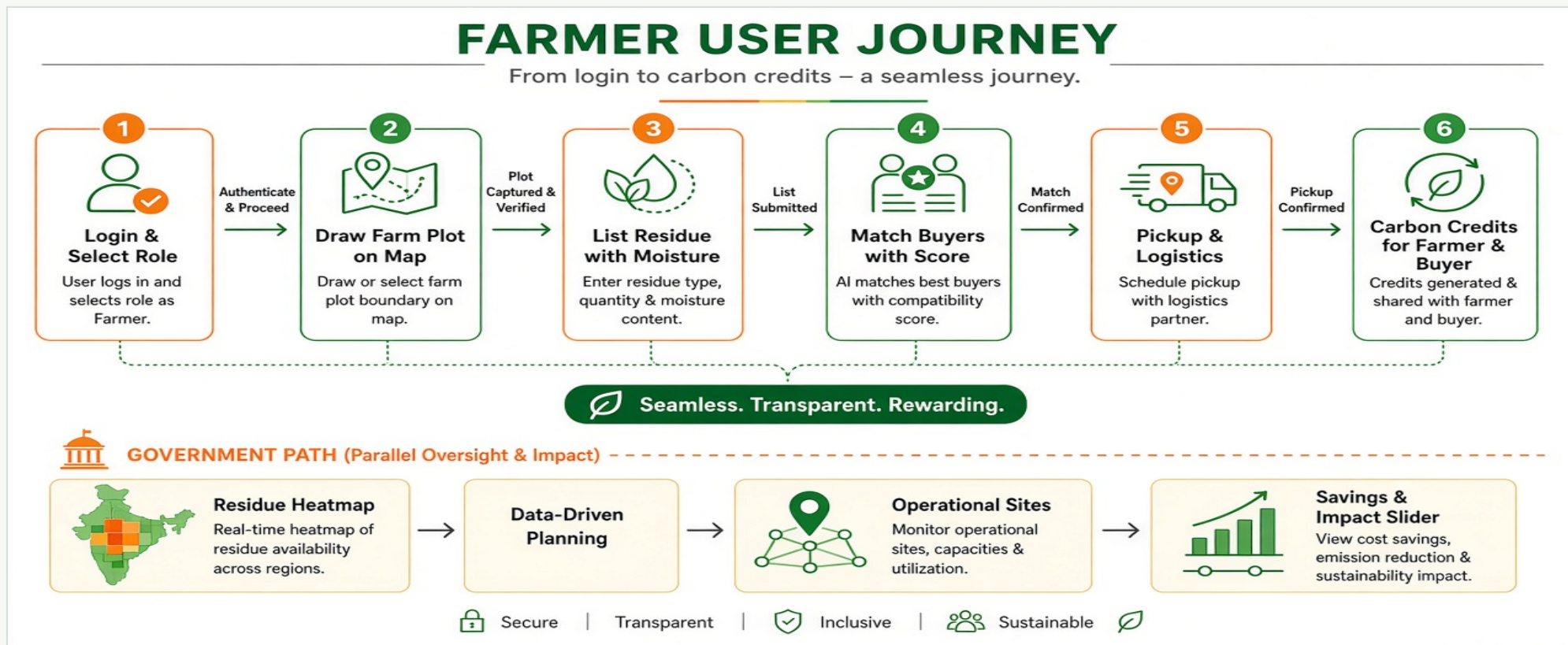


Diagram: farmer path is sequential (role login, map, calculate, list, match, pickup, credits). Government path is parallel - heatmap, operational sites, savings slider - with no farmer PII on that desk.

Seven steps as implemented

1. Role login

Select Farmer / Buyer / Government / Admin. Confirm name, village, district. One session, one desk.

2. Plot map

Click field corners on OSM around Kharar. Sale parcels estimate straw (~2 t/acre rice). Pathways ranked for that tonnage.

3. Burn vs sell

Slide acres, price, moisture, baling cost. See net income vs Rs 0 burn, fine risk, credits, and top Match Score.

4. List residue

Wizard: farm, quality, moisture, bales, expected Rs/t, pathway. Classification can be overridden.

5. Match

Top buyers scored. Moisture above plant spec is penalised. Quantity and radius remain hard filters.

7. Credits

On complete: farmer credits for seed/urea; buyer procurement credits for the tax/CSR panel.

Role-first login


AgriNova
Login — select your role to continue

Role


Farmer
Sell residue · credits


Buyer
Procure biomass


Government
Air & policy desk


Admin
Operations · MRV

Profile

Ramesh Singh — Smallholder · 6 acres

Name

Ramesh Singh

District

Patiala

Village / office

Kharar

Phone

+91 98765 10421

Land (acres)

6

Age

42

Gender

Male

Login as Ramesh

What this screen does

Four roles on one form. Farmer is selected here with a Punjab profile (Kharar, 6 acres). Login is identity for the demo, not a password IdP. Citizen and officer desks stay separate after this step.

Farmer workspace home

AgriNova
Farmer workspace

Dashboard

Plot map

Burn vs sell

Sell residue

My listings

EN

HI

BN

RS Ramesh Singh Farmer

⌵

SIGNED IN

Ramesh Singh
Kharar, Patiala

Dashboard

Plot map

Burn vs sell

Sell residue

My listings

Credits

Kisan AI

Learn farming

Pickup

Demand

Weather

Mandi

Ledger

RS

Ramesh Singh

Kharar, Patiala · 6 acres · rice-wheat · KCC ---4821

"12 days left for wheat. Last year I burned. This year I want the money and no notice from the thana."

Burn vs sell

List residue

Days to wheat sowing

12

PUBLIC DATA

Typical 10-20 day CRM window.

Straw on the field now

5.0 t

2.5 rice acres × ~2 t/acre (TARI-style factor)

Credit wallet

0

Redeem for urea / seed →

This season's job

Sell ~5 t rice straw before the window closes

Last kharif you burned 4.8 t — ₹0 and a fire risk. Moisture, bales, and land area go on the listing so GreenPower can bid.

Start listing →

Map sown plots

⚠ Rice risk — harvest / residue overlap

Harvest early, list residue within 48hrs, book baler immediately

What this screen does

Signed-in farmer only. Sidebar is the residue desk: map, burn vs sell, sell, listings, credits, Kisan AI, learn, pickup. Language toggle EN/HI/BN. Switch account is logout - not a mid-session role flip.

Real map, drawn parcels, straw estimate

 **AgriNova**
Farmer workspace

Dashboard

Plot map

Burn vs sell

Sell residue

My listings

EN HI BN **RS** Ramesh Singh Farmer

SIGNED IN
Ramesh Singh
Kharar, Patiala

Dashboard

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Map your sown plots

Draw sown fields on the real OpenStreetMap around Kharar. Mark parcels for sale — AgriNova estimates straw and ranks pathways.

Crop for next draw

Rice

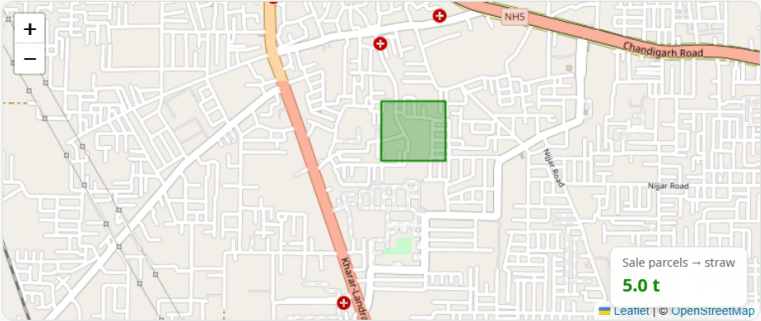
☒ Mark parcel for sale

Clear stroke

Save parcel

Reset parcels

OpenStreetMap - click field corners on the real map around Kharar / Patiala. Acres from geodesic area.



Paddy block A: 2.5 ac · Rice · for sale · ~5.0 t (30.7389, 76.6486)

Best use pathways for ~5.0 t

Biomass power / industry Strong buyer demand and established haul routes.	92
Biochar production Higher price per tonne; suitable for dry straw.	82
Composting	74

What this screen does

OpenStreetMap around Kharar. Green polygon is a sale parcel. 2.5 acres rice maps to ~5.0 t straw. Pathways (biomass, biochar, compost) sit under the map so geography feeds the market.

List residue: farm, quality, match

AgriNova

Farmer workspace

Dashboard

Plot map

Burn vs sell

Sell residue

My listings

EN

HI

BN

RS

Ramesh Singh

Farmer

SIGNED IN

Ramesh Singh

Kharar, Patiala

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Mandi

Ledger

List residue — Ramesh Singh

2.5 acres paddy x ~2 t/acre ≈ 5.0 t straw. Moisture and bales decide whether GreenPower can lift it.

1. farm

2. quality

3. listing

4. pathways

5. offers

6. logistics

7. complete

Farm & land

Farmer

Ramesh Singh

Phone

+91 98765 10421

Location

Kharar, Patiala, Punjab

Total land (acres)

6

Area under this residue (acres)

2.5

Source crop

Rice

Harvest date

01-11-2025

Detected residue

Rice Straw

96% confidence

Rice Straw

Wheat Stubble

Cotton Stalks

Sugarcane Tops

Maize Stover

Continue

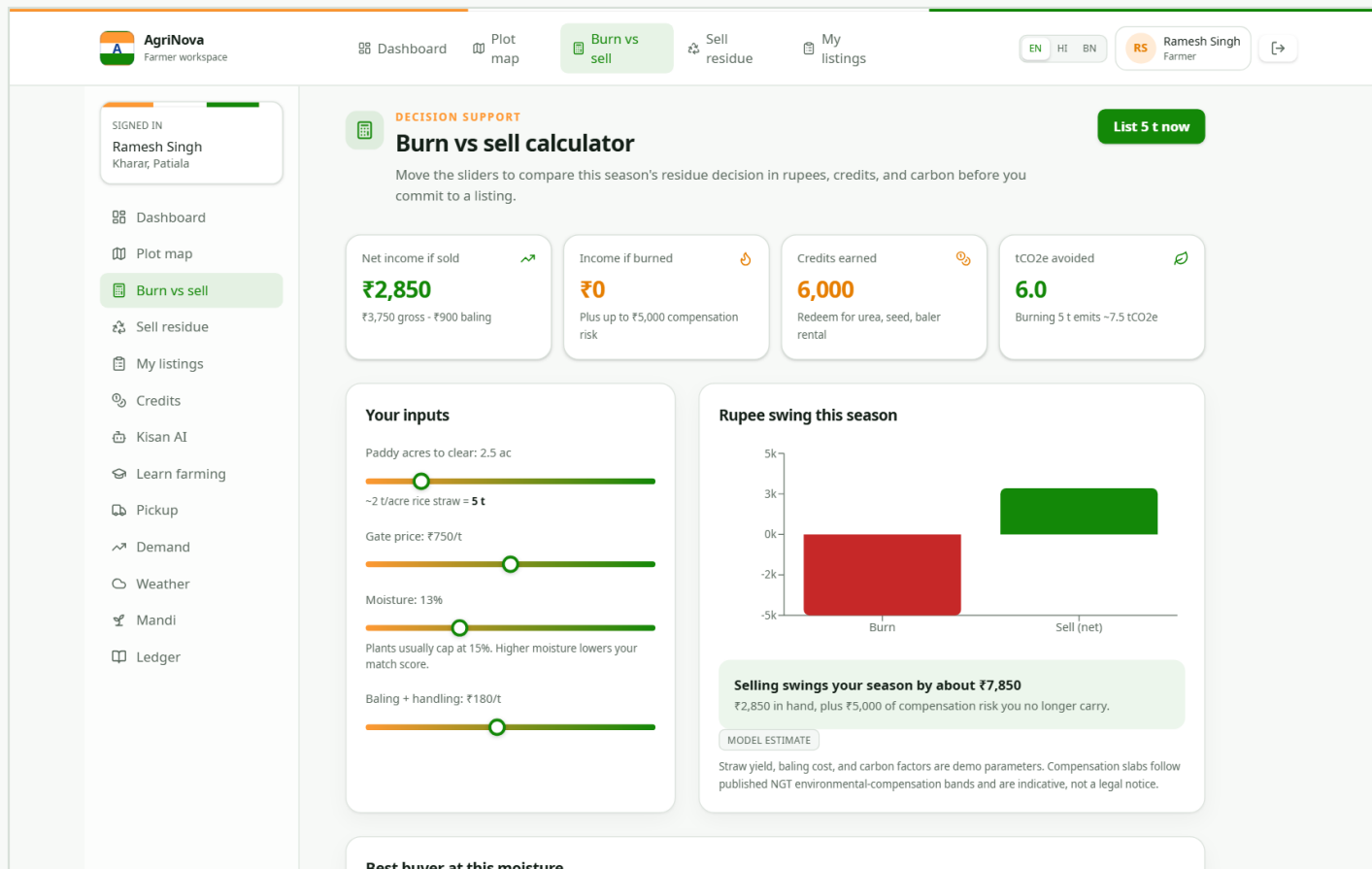
What this screen does

Seven-step golden path. AI classification of Rice Straw with override chips. Moisture later gates whether GreenPower can lift the lot. This is the 5-minute judge path from field to offer.

AgriNova | Crop residue as income | NCSC

11/24

Decision calculator before listing



What this screen does

Interactive sliders for acres, gate price, moisture, and baling cost. Compares net sell income vs burn (Rs 0 + NGT fine risk), credits, avoided tCO₂e, and the best buyer Match Score at that moisture.

Soil monitoring card + OpenAI chat

The screenshot displays the AgriNova Farmer workspace interface. At the top, the header includes the AgriNova logo, navigation links (Dashboard, Plot map, Burn vs sell, Sell residue, My listings), language selectors (EN, HI, BN), and a user profile for Ramesh Singh. A left sidebar lists various features, with 'Kisan AI' highlighted. The main content area is titled 'Kisan AI' and includes a subtitle: 'Tuned for residue, soil, and sowing — uses your soil card and plot map.' Below this, a 'Soil profile' section contains input fields for Soil type (Alluvial), pH (7.2), Organic carbon % (0.45), and Nitrogen status (Medium). A text area for 'Notes for Kisan AI' contains the text: 'Patala belt alluvial — demo soil card for Kisan AI.' At the bottom, a 'KisanSathi' chat interface is shown, featuring a microphone icon, a text input area with the prompt 'Type or tap the mic — Hindi/English. Answers use OpenAI when configured.', and an 'OpenAI live' button.

What this screen does

Editable soil card: alluvial, pH 7.2, OC 0.45%, N medium. KisanSathi uses that card plus mapped acres. Mic and spoken replies. OpenAI live when a local key is set; otherwise researched fallbacks.

New methods taught inside the same desk

Dashboard
Plot map
Burn vs sell
Sell residue
My listings

EN HI BN
RS Ramesh Singh Farmer

SIGNED IN
Ramesh Singh
Kharar, Patiala

Dashboard
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Ledger

Kisan learning

New farming methods

Pick a practice, then ask Kisan AI by text or voice (OpenAI). Hindi + English ok.

All
Residue
Sowing
Soil
Water
Income

Happy Seeder / zero-till wheat
हैप्पी सीडर
Sow wheat into standing rice residue without burning. Saves time between harvest and sowing.

Direct Seeded Rice (DSR)
डीएसआर धान
Rice without puddling transplant. Cuts labour and can save irrigation water when timed well.

Bale & sell residue
फसली बेचना
Bale dry straw (≤15% moisture) and sell via AgriNova instead of burning.

On-farm biochar pathway
बायोचार
Convert part of residue to biochar for soil carbon — slower cash than biomass sale.

Surface mulch & incorporation
मलच / मिट्टी में मिलाना
Leave chopped residue as mulch or incorporate lightly where buyers are far.

Pulse intercrop / diversification
दलहन अंतरफसल
Add short-duration pulses or oilseeds where water and market allow.

Happy Seeder / zero-till wheat

- Avoids residue burn fines
- Keeps soil cover
- Often lower diesel than full tillage

Voice on in chat

Learn: Happy Seeder / zero-till wheat
Ramesh Singh · Patiala

OpenAI live

Type or tap the mic — Hindi/English. Answers use OpenAI when configured.

What this screen does

Practice cards (Happy Seeder, DSR, biochar, mulch, pulses) seed the chatbot. Voice on. The farmer stays in-role - learning is not a separate government portal.

AgriNova | Crop residue as income | NCSC

14/24

Farmer wallet after sales

AgriNova
Farmer workspace

Dashboard

Plot map

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EN HI BN

RS Ramesh Singh Farmer

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Kharar, Patiala

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FARMER WALLET

Carbon credit wallet

Ramesh Singh's credits from completed residue sales and redemptions.

AVAILABLE BALANCE

0

AgriNova Credits - 1 credit = 1 kg CO₂e avoided (demo unit)

Lifetime earned: 0

Redeemed: 0

From ledger: 0 cr.

Redeem for farm inputs

Wheat seed pack (5 kg) 120 cr

PBW-343 variety — demo subsidy via credits

Redeem

Paddy seed (2 kg) 95 cr

Short-duration variety for timely sowing

Redeem

Urea (1 bag) 180 cr

Redeem at partner agro-dealer (demo)

Redeem

DAP (1 bag) 220 cr

Phosphorus for Rabi crops

Redeem

Baler rental (1 day) 350 cr

Shared baler in Patiala cluster

Redeem

Soil health test 80 cr

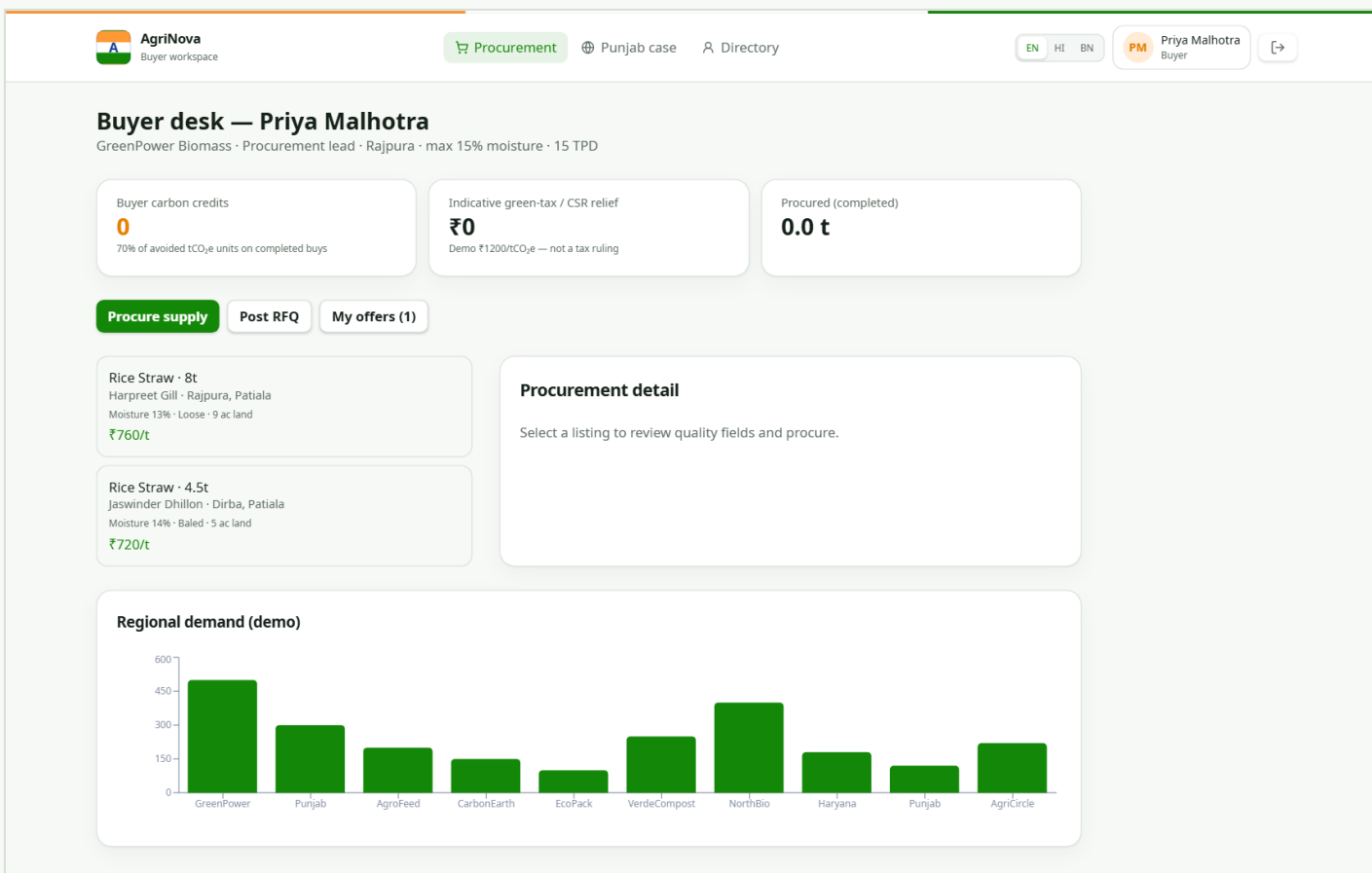
Lab kit via Krishi Vigyan Kendra partner

Redeem

What this screen does

Credits from completed residue sales. Redeem against seed/fertiliser catalogue with toast confirmation. Simran's lived-in sale is already in seed data so this screen is never empty on first load.

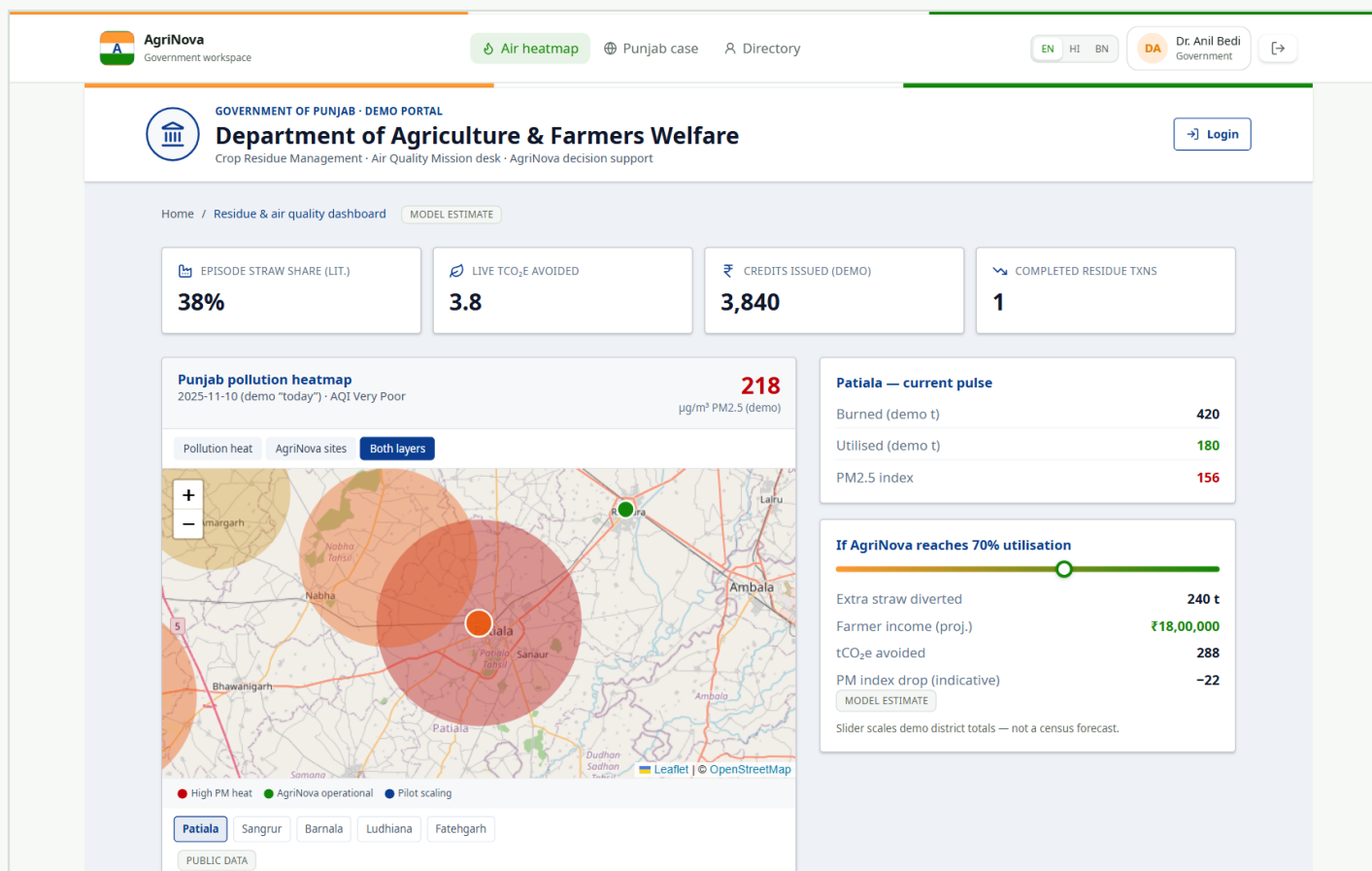
Procurement desk



What this screen does

GreenPower-style inbox: listings with moisture, Rs/t, demand chart. Buyer carbon credits and indicative green-tax/CSR relief. Plant spec (max 15% moisture, 15 TPD) sits in the header.

Agriculture portal: heatmap + savings



What this screen does

Department of Agriculture masthead. Heat blobs = demo PM intensity. Green markers = operational AgriNova clusters. Utilisation slider projects straw diverted, farmer income, and tCO₂e. Model estimate, not a census forecast.

Patiala-Sangrur cluster as the product's home

The run
practice

Cluster

25 farms, ~180 t straw in the pack. Default story:
2.5 acres paddy x ~2 t/acre = ~5 t for one listing.

Places

Kharar (farmer plots), Rajpura (GreenPower plant, 8 km), Sangrur (second farmer), Patiala DAO desk.

Without AgriNova

148 t burned / 32 t utilised in the pack. Farmer income Rs 0. Indicative PM index 172.

With 70% utilisation

54 t burned / 126 t utilised. Farmer income Rs 9.45 lakh. ~151 tCO₂e avoided. PM index 138.

Operational sites

Kharar, Sunam, Rajpura live; Barnala pilot. Rollup: straw diverted, burn avoided, income, tCO₂e on the map.

Honesty layer

Labels: demonstration data, model estimate, public data. No statewide forecast from 25 farms.

Capability list in the live product

Role login

Farmer, buyer, government, admin - separate nav, RoleGate on URLs.

OSM plot map

Draw parcels, geodesic acres, straw estimate, pathway ranking.

Burn vs sell

Interactive calculator: income, fire risk, credits, Match Score.

Sell wizard

Classify, value, moisture, offers, logistics, complete.

Match Score

Five factors + moisture penalty vs plant spec.

Kisan AI

OpenAI, Hindi/English, soil-aware, voice in and out.

Soil card

Type, pH, OC%, N status persisted in session.

Learn farming

Happy Seeder, DSR, biochar, mulch, diversification.

Dual credits

Farmer redeem + buyer procurement / CSR panel.

Gov heatmap

Heat + operational markers + utilisation slider.

Carbon ledger

Avoided tCO₂e entries and MRV queue.

Toast UX

Confirm redemptions and soft warnings without blocking.

Lived-in seed

Prior sale and open RFQ so tables are not empty.

i18n

EN / HI / BN toggle on the shell.

Tricolor UI

White base with saffron and India-green accents.

Burn vs sell - decide before you list

Farmers
sell wiza

Inputs

Paddy acres, gate price Rs/t, moisture %, baling + handling cost. Straw yield uses ~2 t/acre (IARI-style demo factor).

Outputs

Gross and net income, NGT environmental-compensation risk bands, AgriNova credits, avoided tCO₂e, top buyer Match Score.

Action

One click lists the calculated tonnage into the sell wizard so the decision becomes a transaction.

- Example: 2.5 acres -> ~5 t. At Rs 750/t and Rs 180/t baling, net ~Rs 2,850 - vs Rs 0 burn and up to Rs 5,000 compensation risk.
- Moisture above 15% surfaces a drying tip and lowers Match Score so the farmer fixes quality before listing.
- Carbon uses the same avoided-tCO₂e service as the ledger, so booth numbers stay consistent across screens.

What the engineering choices enable

Explainable matching

Judges can open a score breakdown: distance, demand, moisture, price, pathway. Not a black box.

Geography is real

OSM tiles and lat/lng parcels make straw-from-land credible. Heatmap uses district HQ coords, not a fake sketch.

AI is bounded

Role system prompts. Soil and acres injected. Missing key -> fallback copy so the booth still runs.

State is demo-safe

localStorage + seed data. Reset possible. No farmer PII on the government desk.

Honest models

Utilisation slider and before/after charts are labelled MODEL ESTIMATE and scaled to the cluster.

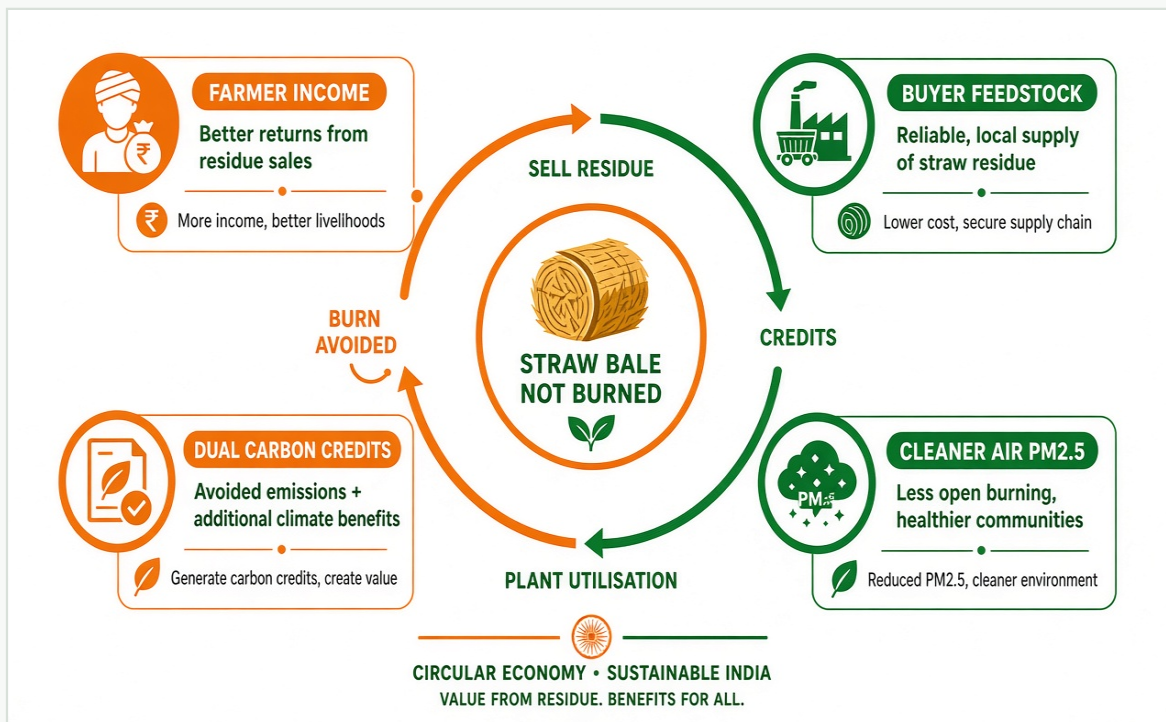
Decision before listing

Burn-vs-sell reuses Match Score and carbon services so the calculator is not a disconnected toy.

Deployed

SPA on Vercel with SPA rewrites. Live URL for judges who will not install Node.

If straw is sold instead of burned



Circular residue loop

Burn avoided feeds utilisation. Plants get feedstock. Farmers get Rs/t plus redeemable credits. Buyers get procurement credits. District PM index falls in the model. The loop is the product: marketplace + credits + officer view.

- Farmer: cash vs Rs 0 burn; credits for seed/fertiliser; calculator shows the swing before listing.
- Buyer: moisture-qualified lots from farms that never reached mill groups.
- Officer: utilised vs burned by district, operational cluster savings, dual-credit totals - aggregates only.
- Climate: avoided tCO2e on completed transactions; 70% utilisation pack ~151 tCO2e in the demo model.

Question the product is designed to test

Can an
10-15 da

Hypothesis

Income + moisture-aware match + pickup logistics + soil-aware advice + visible rupee math beats the burn habit for a 4-6 acre rice-wheat farm.

Method

Interactive simulation. Lived-in seed. Complete a sale in-session. Compare without vs 70% utilisation. West Bengal pack for comparative CRM.

Limits (stated in the UI)

Not cadastral land records. Not live CPCB sensors. Not a certified carbon registry. Buyer tax panel is indicative, not a finance-ministry rule. Fine bands are published NGT reference slabs, not a legal notice. Cluster numbers are not state totals.

AgriNova is live

Open the

<https://agrinova-ochre.vercel.app>

Login -> Farmer -> Plot map -> Burn vs sell -> Sell residue

Login -> Government -> Air heatmap + savings slider

Login -> Farmer -> Kisan AI (soil + voice)

In-app slides: /presentation

PDF download: /AgriNova_Presentation.pdf

Tagline

Predict.

Protect.

Prosper.

Recycle.

National Children's Science Congress | Punjab case study demo | Product as shipped